

ARBOR — Beocat User Guide

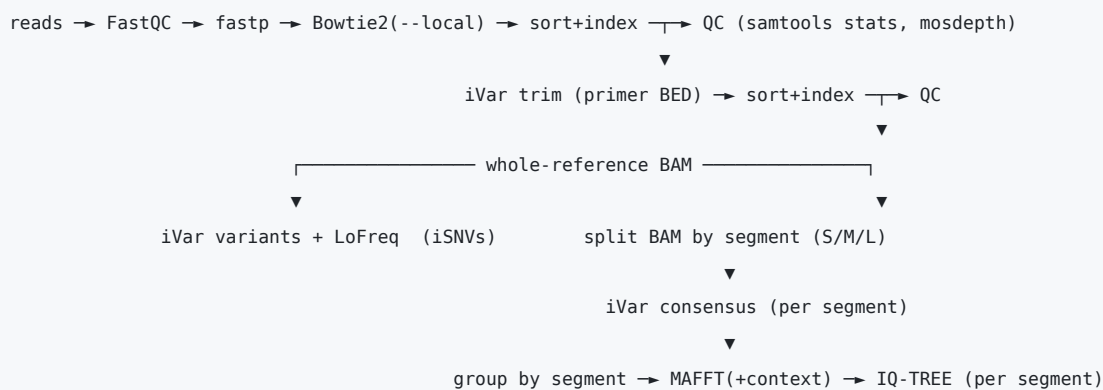
How to run **ARBOR** (viral tiling-amplicon: primer-trimmed mapping → iSNVs → per-segment consensus → phylogeny) on KSU [Beocat](#), from Illumina paired-end `fastq.gz`.

The RVFV MP-12 reference, primer BED, and `--segments` are **bundled defaults** — you only need a samplesheet to run.

Launch everything from a Beocat **login node** (any `icr-*` host, e.g. `icr-helios`). A small head job drives Nextflow, which submits each step as its own SLURM job under the `beocat` profile.

First time on Beocat? Skim [Good to know](#) at the bottom first — it covers the handful of things that surprise new users (where reads go, `Undetermined`, the `TD0ERKS_` label, and copying results off scratch).

What it does



1. Upload your reads to Beocat

Your fastqs are local (BaseSpace/BCLConvert output, named like `R3D720_S5_L001_R1_001.fastq.gz`). Put them on **scratch, not your home dir** — home has a small quota, and `/fastscratch` is the big, fast space meant for job I/O. First make the folder (on Beocat):

```
mkdir -p /fastscratch/$USER/reads
```

Then copy from the machine where the data lives. From **Windows PowerShell** (or use WinSCP / MobaXterm), replacing `<user>`, `<beocat-login>`, and the local path with your own:

```
# If the *.fastq.gz sit directly in the BCLConvert output folder:
scp "C:\path\to\BCLConvert_output\*.fastq.gz" <user>@<beocat-login>:/fastscratch/<user>/reads/

# If they're in per-sample subfolders, flatten them into one reads/ dir instead:
scp -r "C:\path\to\BCLConvert_output\*\*.fastq.gz" <user>@<beocat-login>:/fastscratch/<user>/reads/
```

`<beocat-login>` is the same host you SSH into (e.g. `headnode.beocat.ksu.edu`). BCLConvert's `Undetermined_*` files can come along — the samplesheet builder skips them automatically. End state: `/fastscratch/$USER/reads/` full of `*_R1_001.fastq.gz` / `*_R2_001.fastq.gz`.

2. Get the pipeline + build the samplesheet

```
cd /fastscratch/$USER
git clone https://github.com/tdoerks/ARBOR.git
cd ARBOR/tests/beocat

# build the samplesheet from your reads folder (handles the _S#_L00#_R1_001 naming)
bash make_samplesheet.sh /fastscratch/$USER/reads > samplesheet.csv
column -t -s, samplesheet.csv          # eyeball: one row per sample, R1 + R2 filled
```

The generator derives the sample name as everything before `_S<n>` (so `R3D720_S5_L001_R1_001.fastq.gz` → `R3D720`). If a sample spans **multiple lanes** (L001 and L002), it warns and uses the first lane — merge lanes beforehand if you need both.

3. Launch

Optional — **email notifications**: edit `run_arbor.sbatch` and uncomment the two `##SBATCH --mail-*` lines, setting `--mail-user` to your own KSU address. Left off by default. Then:

```
sbatch run_arbor.sbatch
```

That's it — reference, primers, and segment names are bundled defaults. The head job runs your **local clone** (so a `git pull` always takes effect) under the `beocat` profile, roughly equivalent to:

```
nextflow run . -profile beocat --input samplesheet.csv --outdir results -resume
```

4. Monitor

```
squeue -u $USER          # head job + the per-step jobs it spawns
tail -f arbor_head_*.log  # live Nextflow progress
```

5. Results (`results/`)

Path	What
<code>results/multiqc/multiqc_report.html</code>	combined QC (FastQC, fastp, samtools stats, mosdepth)
<code>results/mosdepth/</code>	per-sample coverage — check for amplicon "valleys" (failed primer pairs)
<code>results/ivar/</code>	iVar variant tables (<code>.tsv</code>) — frequency-based iSNVs
<code>results/lofreq/</code>	LoFreq VCFs — significance-tested low-frequency variants
<code>results/ivarconsensus/</code> (or <code>ivar/</code>)	per-segment consensus FASTA per sample
<code>results/iqtree/*.treefile</code>	one core-genome tree per segment (S, M, L)

Defaults you can override

Flag	Default	Notes
<code>--reference</code>	bundled RVFV MP-12	swap for another virus
<code>--primer_bed</code>	bundled <code>rvfv_amplicons_v4.bed</code>	your tiling scheme
<code>--segments</code>	<code>NC_014395_S,NC_014396_M,NC_014397_L</code>	must match reference FASTA headers
<code>--context_fasta</code>	none	external NCBI strains to fold into the trees
<code>--ivar_variants_min_freq / _min_depth / _min_qual</code>	0.01 / 250 / 20	iVar iSNV thresholds
<code>--lofreq_sig / --lofreq_min_cov</code>	0.005 / 100	LoFreq thresholds
<code>--consensus_min_depth</code>	10	below this → N
<code>--iqtree_bootstrap / --iqtree_model</code>	1000 / MFP	phylogeny
<code>--skip_lofreq / --skip_ivar_variants / --skip_phylogeny</code>	false	drop a stage

Troubleshooting

nextflow / singularity not found — Beocat module names differ; find them with `module spider nextflow / module spider singularity` and update the two `module load` lines in `run_arbor.sbatch`.

Network errors on first run — container images download on first use. If compute nodes have no internet, pre-pull on a login node (ask and we'll set up `NXF_SINGULARITY_CACHEDIR` staging).

Empty/low coverage on a segment — check `results/mosdepth/`; coverage valleys indicate a failed primer pair in that amplicon. The trees still build per segment from whatever consensus is recovered.

Re-running — `-resume` is on; just `sbatch run_arbor.sbatch` again — completed steps are cached.

Good to know (first-timers)

Things that surprise people on their first ARBOR run — none are errors:

- **Reads go on `/fastscratch/$USER/`, not your home dir.** Home has a small quota; scratch is the fast, roomy space for job I/O (work dir, container cache, and `results/` all live there too).
- **`/fastscratch` is auto-purged** after a period of inactivity. When the run finishes, **copy `results/` somewhere permanent** (home, lab storage) — scratch is for active runs, not keeping.
- **Undetermined_* fastqs are skipped automatically** — you'll see `INFO: skipping Undetermined_...`. That's expected (those are unassigned barcodes, not a sample).
- **The process list shows `TDOERKS_ARBOR:ARBOR:...`**. That `TDOERKS_` prefix is just the pipeline's *name* (from `manifest.name = 'tdoerks/arbor'` — the GitHub repo), like nf-core pipelines show `NFCORE_...`. It is **not** an account or a path: your job runs as **you** and writes under **your** `/fastscratch/$USER/`. Verify any time with `echo $USER`, `pwd`, and `queue -u $USER`.
- **The first run is quiet for several minutes.** The head job installs Nextflow and pulls the Singularity containers before any step starts — the log will sit still, then fill in. Later runs reuse the cache and start fast.
- **queue drip-feeds jobs.** The `beocat` profile throttles submission (≤ 100 queued, $\leq 10/\text{min}$) so a big run doesn't flood the scheduler — you won't see all tasks at once. Normal.
- **Only per-user edit needed** is the optional email in `run_arbor.sbatch`; everything else keys off `$USER`, so the scripts work for anyone without renaming.

Explore the results

The dashboard at [dashboard/arbor_dashboard.html](#) opens in any browser (no install) — drag in the run's `ivar/*.tsv` , `lofreq/*.vcf` , `mosdepth/*` , `iqtree/*.treefile` (and optionally `pipeline_info/params.json`) to browse variants, coverage, trees, and the parameters used.